Problem Statement

xyz co's 2014-2018 sales data to identify key revenue and profit drivers across products,channels and regions,uncover seasonal trends and outliers, and align performance against budgets. Use these insights to optimize pricing, promotions and market expansion for sustainable growth and reduced concentration risk.

In [1]:
```
# importing libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (10,6)
```

In [2]:
```
sheets = pd.read_excel('regional_sales.xlsx', sheet_name=None)
sheets
```

Out[2]:
```
{'Sales Orders':         OrderNumber  OrderDate  Customer Name Index      Chann
el  \
 0       SO - 000225 2014-01-01                 126    Wholesale
 1      SO - 0003378 2014-01-01                  96  Distributor
 2      SO - 0005126 2014-01-01                   8    Wholesale
 3      SO - 0005614 2014-01-01                  42       Export
 4      SO - 0005781 2014-01-01                  73    Wholesale
 ...             ...        ...                 ...          ...
 64099  SO - 0007573 2018-02-28                  74    Wholesale
 64100  SO - 0007706 2018-02-28                  51       Export
 64101  SO - 0007718 2018-02-28                 136  Distributor
 64102  SO - 0008084 2018-02-28                 158  Distributor
 64103  SO - 0008654 2018-02-28                  22  Distributor

       Currency Code Warehouse Code  Delivery Region Index  \
 0               USD         AXW291                     364
 1               USD         AXW291                     488
 2               USD         AXW291                     155
 3               USD         AXW291                     473
 4               USD         AXW291                     256
 ...             ...            ...                     ...
 64099           USD         AXW291                     825
 64100           USD         NXH382                     444
 64101           USD         AXW291                     312
 64102           USD         AXW291                     737
 64103           USD         AXW291                     807

        Product Description Index  Order Quantity  Unit Price  Line Total  \
 0                             27               6      2499.1     14994.6
 1                             20              11      2351.7     25868.7
 2                             26               6       978.2      5869.2
 3                              7               7      2338.3     16368.1
 4                              8               8      2291.4     18331.2
 ...                          ...             ...         ...         ...
 64099                         26              12      1815.7     21788.4
 64100                         21               6       864.3      5185.8
 64101                         13              11      3953.0     43483.0
 64102                         20               7      3959.7     27717.9
 64103                         15               8       998.3      7986.4

        Total Unit Cost
 0             1824.343
 1             1269.918
 2              684.740
 3             1028.852
 4             1260.270
 ...                ...
 64099          980.478
 64100          579.081
 64101         2648.510
 64102         2930.178
 64103          848.555

 [64104 rows x 12 columns],
 'Customers':      Customer Index Customer Names
 0                 1  Geiss Company
 1                 2  Jaxbean Group
 2                 3     Ascend Ltd
 3                 4      Eire Corp
 4                 5   Blogtags Ltd
```

```
..              ...            ...
170             171   Accord Group
171             172   BB17 Company
172             173          H Ltd
173             174   Tekfly Group
174             175   SHISEIDO Ltd

[175 rows x 2 columns],
'Regions':       id        name                                  county state
_code  \
0      1      Auburn                                 Lee County         AL
1      2  Birmingham           Shelby County/Jefferson County         AL
2      3     Decatur           Limestone County/Morgan County         AL
3      4      Dothan  Dale County/Houston County/Henry County         AL
4      5      Hoover           Shelby County/Jefferson County         AL
..   ...         ...                                       ...        ...
989  990      Racine                              Racine County         WI
990  991    Waukesha                            Waukesha County         WI
991  992  West Allis                           Milwaukee County         WI
992  993      Casper                             Natrona County         WY
993  994    Cheyenne                             Laramie County         WY

         state  type  latitude  longitude  area_code  population  households
\
0      Alabama  City  32.60986  -85.48078        334       62059       21767
1      Alabama  City  33.52744  -86.79905        205      212461       89972
2      Alabama  City  34.57332  -86.99214        256       55437       22294
3      Alabama  City  31.23370  -85.40682        334       68567       25913
4      Alabama  City  33.37695  -86.80558        205       84848       32789
..         ...   ...       ...        ...        ...         ...         ...
989  Wisconsin  City  42.72613  -87.78285        262       77742       29851
990  Wisconsin  City  43.01168  -88.23148        262       71970       28331
991  Wisconsin  City  43.01668  -88.00703        414       60620       27457
992    Wyoming  City  42.86663 -106.31308        307       60285       23814
993    Wyoming  City  41.13998 -104.82025        307       63335       25130

     median_income  land_area  water_area        time_zone
0            38342  152375113     2646161  America/Chicago
1            31061  378353942     6591013  America/Chicago
2            41496  141006257    17594716  America/Chicago
3            42426  232166237      835468  America/Chicago
4            77146  122016784     2553332  America/Chicago
..             ...        ...         ...              ...
989          41455   40111757     8282116  America/Chicago
990          59547   65856921      700602  America/Chicago
991          45221   29480532       60789  America/Chicago
992          57790   68507280      911081   America/Denver
993          58219   69964749      282335   America/Denver

[994 rows x 15 columns],
'State Regions':        Column1               Column2    Column3
0   State Code                 State     Region
1           AL               Alabama      South
2           AR              Arkansas      South
3           AZ               Arizona       West
4           CA            California       West
5           CO              Colorado       West
6           CT           Connecticut  Northeast
7           DC  District of Columbia      South
8           DE              Delaware      South
```

```
9           FL               Florida      South
10          GA               Georgia      South
11          HI                Hawaii       West
12          IA                  Iowa    Midwest
13          ID                 Idaho       West
14          IL              Illinois    Midwest
15          IN               Indiana    Midwest
16          KS                Kansas    Midwest
17          KY              Kentucky      South
18          LA             Louisiana      South
19          MA         Massachusetts  Northeast
20          MD              Maryland      South
21          ME                 Maine  Northeast
22          MI              Michigan    Midwest
23          MN             Minnesota    Midwest
24          MO              Missouri    Midwest
25          MS           Mississippi      South
26          MT               Montana       West
27          NC        North Carolina      South
28          ND          North Dakota    Midwest
29          NE              Nebraska    Midwest
30          NH         New Hampshire  Northeast
31          NJ            New Jersey  Northeast
32          NM            New Mexico       West
33          NV                Nevada       West
34          NY              New York  Northeast
35          OH                  Ohio    Midwest
36          OK              Oklahoma      South
37          OR                Oregon       West
38          PA          Pennsylvania  Northeast
39          RI          Rhode Island  Northeast
40          SC        South Carolina      South
41          SD          South Dakota    Midwest
42          TN             Tennessee      South
43          TX                 Texas      South
44          UT                  Utah       West
45          VA              Virginia      South
46          WA            Washington       West
47          WI             Wisconsin    Midwest
48          WY               Wyoming       West,
'Products':     Index Product Name
0       1    Product 1
1       2    Product 2
2       3    Product 3
3       4    Product 4
4       5    Product 5
5       6    Product 6
6       7    Product 7
7       8    Product 8
8       9    Product 9
9      10   Product 10
10     11   Product 11
11     12   Product 12
12     13   Product 13
13     14   Product 14
14     15   Product 15
15     16   Product 16
16     17   Product 17
17     18   Product 18
18     19   Product 19
```

```
19     20   Product 20
20     21   Product 21
21     22   Product 22
22     23   Product 23
23     24   Product 24
24     25   Product 25
25     26   Product 26
26     27   Product 27
27     28   Product 28
28     29   Product 29
29     30   Product 30,
'2017 Budgets':    Product Name  2017 Budgets
0     Product 1   3016489.209
1     Product 2   3050087.565
2     Product 3   2642352.432
3     Product 4   2885560.824
4     Product 5   3925424.542
5     Product 6   1672229.416
6     Product 7    889737.555
7     Product 8   1085037.329
8     Product 9    710415.405
9    Product 10    864331.624
10   Product 11   1230458.484
11   Product 12   1356976.996
12   Product 13   3720639.645
13   Product 14   4028698.409
14   Product 15   3299478.315
15   Product 16   2062061.144
16   Product 17   2119616.824
17   Product 18   2309499.045
18   Product 19   1645841.600
19   Product 20   2067108.120
20   Product 21    959899.218
21   Product 22    663601.902
22   Product 23   1135815.758
23   Product 24    756557.970
24   Product 25   5342315.484
25   Product 26   5685138.270
26   Product 27    964940.231
27   Product 28   1003867.767
28   Product 29    594471.570
29   Product 30   1011609.684}
```

In [3]:
```
# Assign dataframes to each sheets
df_sales = sheets['Sales Orders']
df_customers = sheets['Customers']
df_products = sheets['Products']
df_regions = sheets['Regions']
df_state_reg = sheets['State Regions']
df_budgets = sheets['2017 Budgets']
```

In [4]:
```
df_sales.head()
```

Out[4]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

In [5]:
```
df_sales.shape
```

Out[5]:
```
(64104, 12)
```

In [6]:
```
print('df_sales shape:', df_sales.shape)
print('df_customers shape:', df_customers.shape)
print('df_products shape:', df_products.shape)
print('df_regions shape:', df_regions.shape)
print('df_state_reg shape:', df_state_reg.shape)
print('df_budgets shape:', df_budgets.shape)
```

```
df_sales shape: (64104, 12)
df_customers shape: (175, 2)
df_products shape: (30, 2)
df_regions shape: (994, 15)
df_state_reg shape: (49, 3)
df_budgets shape: (30, 2)
```

In [7]:
```
df_sales.head()
```

Out[7]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

In [8]:
```
df_customers.head()
```

Out[8]:

	Customer Index	Customer Names
0	1	Geiss Company
1	2	Jaxbean Group
2	3	Ascend Ltd
3	4	Eire Corp
4	5	Blogtags Ltd

In [9]:
```
df_products.head()
```

Out[9]:

	Index	Product Name
0	1	Product 1
1	2	Product 2
2	3	Product 3
3	4	Product 4
4	5	Product 5

In [10]:
```
df_budgets.head()
```

Out[10]:

	Product Name	2017 Budgets
0	Product 1	3016489.209
1	Product 2	3050087.565
2	Product 3	2642352.432
3	Product 4	2885560.824
4	Product 5	3925424.542

In [11]:
```
df_state_reg.head()
```

Out[11]:

	Column1	Column2	Column3
0	State Code	State	Region
1	AL	Alabama	South
2	AR	Arkansas	South
3	AZ	Arizona	West
4	CA	California	West

In [12]:
```
df_regions.head()
```

Out[12]:

	id	name	county	state_code	state	type	latitude	longitude	a
0	1	Auburn	Lee County	AL	Alabama	City	32.60986	-85.48078	
1	2	Birmingham	Shelby County/Jefferson County	AL	Alabama	City	33.52744	-86.79905	
2	3	Decatur	Limestone County/Morgan County	AL	Alabama	City	34.57332	-86.99214	
3	4	Dothan	Dale County/Houston County/Henry County	AL	Alabama	City	31.23370	-85.40682	
4	5	Hoover	Shelby County/Jefferson County	AL	Alabama	City	33.37695	-86.80558	

In [13]:

```
new_header = df_state_reg.iloc[0]  # grab the first row for the header
df_state_reg.columns = new_header  # set the header row as the df header
df_state_reg = df_state_reg[1:].reset_index(
    drop=True)  # take the data less the header row
df_state_reg.head(5)
```

Out[13]:

	State Code	State	Region
0	AL	Alabama	South
1	AR	Arkansas	South
2	AZ	Arizona	West
3	CA	California	West
4	CO	Colorado	West

In [14]:

```
df_sales.isnull().sum()
```

Out[14]:

```
OrderNumber                  0
OrderDate                    0
Customer Name Index          0
Channel                      0
Currency Code                0
Warehouse Code               0
Delivery Region Index        0
Product Description Index    0
Order Quantity               0
Unit Price                   0
Line Total                   0
Total Unit Cost              0
dtype: int64
```

In [15]:

```
df_products.isnull().sum()
```

Out[15]:
```
Index           0
Product Name    0
dtype: int64
```

In [16]:
```
df_regions.isnull().sum()
```

Out[16]:
```
id               0
name             0
county           0
state_code       0
state            0
type             0
latitude         0
longitude        0
area_code        0
population       0
households       0
median_income    0
land_area        0
water_area       0
time_zone        0
dtype: int64
```

In [17]:
```
df_budgets.isnull().sum()
```

Out[17]:
```
Product Name    0
2017 Budgets    0
dtype: int64
```

data cleaning and wrangling

In [18]:
```
# Merge with customers
df = df_sales.merge(df_customers,
                    how='left',
                    left_on='Customer Name Index',
                    right_on='Customer Index'
                    )
```

In [19]:
```
df.head()
```

Out[19]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

In [20]:

```
# Merge with products
df = df.merge(df_products,
              how='left',
              left_on='Product Description Index',
              right_on='Index')
```

In [21]:

```
df.head(5)
```

Out[21]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

In [22]:

```
# Merge with regions
df = df.merge(df_regions,
              how='left',
              left_on='Delivery Region Index',
              right_on='id')
```

In [23]:

```
df.head(1)
```

Out[23]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364

1 rows × 31 columns

In [24]:
```
df = df.merge(df_state_reg[["State Code", "Region"]],
              how='left',
              left_on='state_code',
              right_on='State Code')
```

In [25]:
```
# Merge with budgets
df = df.merge(df_budgets,
              how='left',
              on='Product Name')
```

In [26]:
```
df.head()
```

Out[26]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

5 rows × 34 columns

In [27]:
```
df.to_csv('files.csv')
```

In [28]:
```
# Cleaning up redudnant columns
cols_to_drop = ['Customer Index', 'Index', 'id', 'State Code']
df = df.drop(columns=cols_to_drop, errors='ignore')
```

In [29]:
```
df.head()
```

Out[29]:

	OrderNumber	OrderDate	Customer Name Index	Channel	Currency Code	Warehouse Code	Delivery Region Index
0	SO - 000225	2014-01-01	126	Wholesale	USD	AXW291	364
1	SO - 0003378	2014-01-01	96	Distributor	USD	AXW291	488
2	SO - 0005126	2014-01-01	8	Wholesale	USD	AXW291	155
3	SO - 0005614	2014-01-01	42	Export	USD	AXW291	473
4	SO - 0005781	2014-01-01	73	Wholesale	USD	AXW291	256

5 rows × 30 columns

In [30]:
```
# convert all columns to lowercase for ease of use
df.columns = df.columns.str.lower()
df.columns.values
```

Out[30]:
```
array(['ordernumber', 'orderdate', 'customer name index', 'channel',
       'currency code', 'warehouse code', 'delivery region index',
       'product description index', 'order quantity', 'unit price',
       'line total', 'total unit cost', 'customer names', 'product name',
       'name', 'county', 'state_code', 'state', 'type', 'latitude',
       'longitude', 'area_code', 'population', 'households',
       'median_income', 'land_area', 'water_area', 'time_zone', 'region',
       '2017 budgets'], dtype=object)
```

In [31]:
```
# keeping the important columns for analysis
cols_to_keep = ['ordernumber',
                'orderdate',
                'customer names',
                'channel',
                'product name',
                'order quantity',
                'unit price',
                'line total',
                'total unit cost',
                'state_code',
                'county',
                'state',
                'region',
                'latitude',
                'longitude',
                'median_income',
                '2017 budgets']
```

In [32]:
```
df = df[cols_to_keep]
```

In [33]:
```
df.head()
```

Out[33]:

	ordernumber	orderdate	customer names	channel	product name	order quantity	unit price	lin tota
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	6	2499.1	14994.
1	SO - 0003378	2014-01-01	Thoughtmix Ltd	Distributor	Product 20	11	2351.7	25868.
2	SO - 0005126	2014-01-01	Amerisourc Corp	Wholesale	Product 26	6	978.2	5869.
3	SO - 0005614	2014-01-01	Colgate-Pa Group	Export	Product 7	7	2338.3	16368.
4	SO - 0005781	2014-01-01	Deseret Group	Wholesale	Product 8	8	2291.4	18331.

In [34]:

```
# Rename the columns

df = df.rename(columns={'ordernumber': 'order_number',
                        'orderdate': 'order_date',
                        'customer names': 'customer_name',
                        'product name': 'product_name',
                        'order quantity': 'order_quantity',
                        'unit price': 'unit_price',
                        'line total': 'revenue',
                        'total unit cost': 'total_unit_cost',
                        '2017 budgets': 'budget'})
df.head()
```

Out[34]:

	order_number	order_date	customer_name	channel	product_name	order_quan
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	
1	SO - 0003378	2014-01-01	Thoughtmix Ltd	Distributor	Product 20	
2	SO - 0005126	2014-01-01	Amerisourc Corp	Wholesale	Product 26	
3	SO - 0005614	2014-01-01	Colgate-Pa Group	Export	Product 7	
4	SO - 0005781	2014-01-01	Deseret Group	Wholesale	Product 8	

In [35]:

```
# Blank out budgets for non-2017 orders
df.loc[df['order_date'].dt.year != 2017, 'budget'] = pd.NA
# Line total is revenue
df[['order_date', 'product_name', 'revenue', 'budget']].head()
```

Out[35]:

	order_date	product_name	revenue	budget
0	2014-01-01	Product 27	14994.6	NaN
1	2014-01-01	Product 20	25868.7	NaN
2	2014-01-01	Product 26	5869.2	NaN
3	2014-01-01	Product 7	16368.1	NaN
4	2014-01-01	Product 8	18331.2	NaN

In [36]:
```
df
```

Out[36]:

	order_number	order_date	customer_name	channel	product_name	order_
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	
1	SO - 0003378	2014-01-01	Thoughtmix Ltd	Distributor	Product 20	
2	SO - 0005126	2014-01-01	Amerisourc Corp	Wholesale	Product 26	
3	SO - 0005614	2014-01-01	Colgate-Pa Group	Export	Product 7	
4	SO - 0005781	2014-01-01	Deseret Group	Wholesale	Product 8	
...	...	...	...	...	...	
64099	SO - 0007573	2018-02-28	Dazzlesphe Corp	Wholesale	Product 26	
64100	SO - 0007706	2018-02-28	Yombu Corp	Export	Product 21	
64101	SO - 0007718	2018-02-28	Bath Group	Distributor	Product 13	
64102	SO - 0008084	2018-02-28	Linklinks Ltd	Distributor	Product 20	
64103	SO - 0008654	2018-02-28	SAFEWAY Ltd	Distributor	Product 15	

64104 rows × 17 columns

In [37]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 64104 entries, 0 to 64103
Data columns (total 17 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   order_number     64104 non-null  object
 1   order_date       64104 non-null  datetime64[ns]
 2   customer_name    64104 non-null  object
 3   channel          64104 non-null  object
 4   product_name     64104 non-null  object
 5   order_quantity   64104 non-null  int64
 6   unit_price       64104 non-null  float64
 7   revenue          64104 non-null  float64
 8   total_unit_cost  64104 non-null  float64
 9   state_code       64104 non-null  object
 10  county           64104 non-null  object
 11  state            64104 non-null  object
 12  region           64104 non-null  object
 13  latitude         64104 non-null  float64
 14  longitude        64104 non-null  float64
 15  median_income    64104 non-null  int64
 16  budget           15263 non-null  float64
dtypes: datetime64[ns](1), float64(6), int64(2), object(8)
memory usage: 8.3+ MB
```

In [38]:
```
df.describe()
```

Out[38]:

	order_date	order_quantity	unit_price	revenue	total_unit_cost
count	64104	64104.000000	64104.000000	64104.000000	64104.000000
mean	2016-01-29 01:28:20.935979008	8.441689	2284.380803	19280.682937	1432.083899
min	2014-01-01 00:00:00	5.000000	167.500000	837.500000	68.675000
25%	2015-01-13 00:00:00	6.000000	1031.800000	8019.900000	606.216000
50%	2016-01-27 00:00:00	8.000000	1855.900000	14023.100000	1084.495500
75%	2017-02-13 00:00:00	10.000000	3606.275000	27416.400000	2046.933750
max	2018-02-28 00:00:00	12.000000	6566.000000	78711.600000	5498.556000
std	NaN	2.276217	1663.598141	15429.602766	1107.705728

In [39]:
```
# Filter the dataset to only include orders from 2017 for the EDA, since the bud
df_2017 = df[df['order_date'].dt.year == 2017]
```

In [40]:
```
df.isnull().sum()
```

Out[40]:
```
order_number           0
order_date             0
customer_name          0
channel                0
product_name           0
order_quantity         0
unit_price             0
revenue                0
total_unit_cost        0
state_code             0
county                 0
state                  0
region                 0
latitude               0
longitude              0
median_income          0
budget             48841
dtype: int64
```

In [41]:
```
df_2017.isnull().sum()
```

Out[41]:
```
order_number       0
order_date         0
customer_name      0
channel            0
product_name       0
order_quantity     0
unit_price         0
revenue            0
total_unit_cost    0
state_code         0
county             0
state              0
region             0
latitude           0
longitude          0
median_income      0
budget             0
dtype: int64
```

In [42]:
```
df_2017.head()
```

Out[42]:

	order_number	order_date	customer_name	channel	product_name	order_
46363	SO - 0002544	2017-01-01	NCS Group	Wholesale	Product 30	
46364	SO - 0006431	2017-01-01	Epic Group	Wholesale	Product 13	
46365	SO - 0007491	2017-01-01	State Ltd	Wholesale	Product 15	
46366	SO - 0008741	2017-01-01	Fivebridge Ltd	Wholesale	Product 8	
46367	SO - 0009295	2017-01-01	Tagfeed Ltd	Wholesale	Product 2	

Feature Engineering

In [43]:
```
df['total_cost'] = df['order_quantity'] * df['total_unit_cost']
```

In [44]:
```
df['profit'] = df['revenue'] - df['total_cost']
df['profit_margin_pct'] = df['profit'] / df['revenue'] * 100
df['order_month_name'] = df['order_date'].dt.month_name()
df['order_month_num'] = df['order_date'].dt.month
df.head()
```

Out[44]:

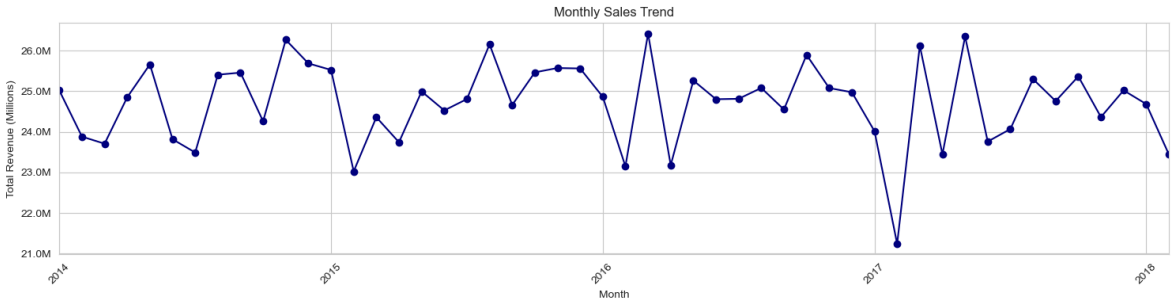
	order_number	order_date	customer_name	channel	product_name	order_quan
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	
1	SO - 0003378	2014-01-01	Thoughtmix Ltd	Distributor	Product 20	
2	SO - 0005126	2014-01-01	Amerisourc Corp	Wholesale	Product 26	
3	SO - 0005614	2014-01-01	Colgate-Pa Group	Export	Product 7	
4	SO - 0005781	2014-01-01	Deseret Group	Wholesale	Product 8	

5 rows × 22 columns

EDA

In [45]:
```
# Convert order_date to monthly period (e.g., 2022-01, 2022-02, etc.)
from matplotlib.ticker import FuncFormatter
df['order_month'] = df['order_date'].dt.to_period('M')

# Calculate total revenue for each month
monthly_sales = df.groupby('order_month')['revenue'].sum()

# Set figure size for clarity
plt.figure(figsize=(15, 4))
```

```
# Plot the monthly sales trend circle markers and navy line
monthly_sales.plot(marker='o', color='navy')

# Scale y-axis values to millions for better readability
formatter = FuncFormatter(lambda x, pos: f'{x/1e6:.1f}M')
plt.gca().yaxis.set_major_formatter(formatter)

# Add title and axis labels
plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Total Revenue (Millions)')

# Rotate x-axis labels for better readability
plt.xticks(rotation=45)

# Adjust layout to prevent clipping of labels
plt.tight_layout()
plt.show()
```



Insights from the figure ---

Sales consistently cycle between $24M and 26$ M, with clear peaks in late spring to early summer (May-June) and troughs each january.

The overall trend remains stable y-o-y, reflecting a reliable seasonal demand pattern.

However, the sharp revenue drop in early 2017 stands out as an outlier, warranting closer investigation into potential causes sucha as market disruption or mistimed promotions.

2.*Monthly Sales Trend (All year Combined)*

Goal: Highlight overall seasonality and patterns by aggregating sales across all years for each calender month

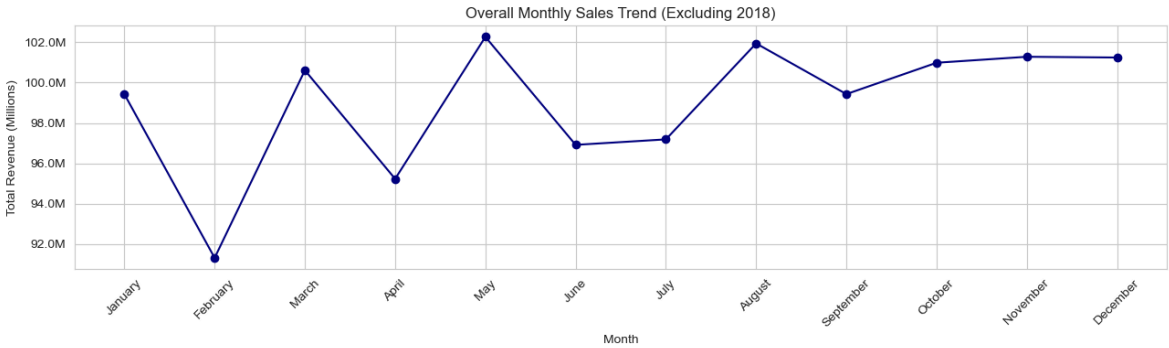
Chart: Line Chart

EDA Typer: Temporal (time series)

Structure: Line with Markers, months ordered January to December based on month number

In [46]:
```
# 1. Filter out and 2018 orders
from matplotlib.ticker import FuncFormatter
df_ = df[df['order_date'].dt.year != 2018]
```

```
# 2. Group by month number and month name, sum revenue, then sort by month numbe
monthly_sales = df_.groupby(['order_month_num', 'order_month_name'])[
    'revenue'].sum().sort_index()

# 3. Plot setup

plt.figure(figsize=(13, 4))
plt.plot(
    monthly_sales.index.get_level_values(1),
    monthly_sales.values,
    marker='o',
    color='navy'
)

# 4. Scale y-axis values to millions for readability
formatter = FuncFormatter(lambda x, pos: f'{x/1e6:.1f}M')
plt.gca().yaxis.set_major_formatter(formatter)

# 5. Add title and axis labels
plt.title('Overall Monthly Sales Trend (Excluding 2018)')
plt.xlabel('Month')
plt.ylabel('Total Revenue (Millions)')

# 6. Rotate x-axis labels for readability
plt.xticks(rotation=45)

# 7. Adjust layout to prevent clipping
plt.tight_layout()

# 8. Display the plot
plt.show()
```



Insights ---

Across all years, January begins strong with roughly $99M, followedbyasteepdeclinethroughApril'slowpoint($ 95 M).

Sales rebound in May and August (~$102M)beforesettingintoaplateauof$99-101 M form September to December.

This pattern reveals a strong post-New year surge, a spring dip, and a mid-summer bump each calender year.

3.Top 10 Products by Revenue (in Millions)

Goal: Identify the highest-grossing products to focus marketing and inventory efforts

Chart:Horizontal bar chart

EDA Type:Univariate

Structure:Bars sorted descending to show top 10 prodcuts with revenue scaled in millions

In [47]:
```
# Calculte total revenue for each product and convert values to millions for bet
top_prod = df.groupby('product_name')['revenue'].sum()/1_000_000

# Select the top 10 products by revenue
top_prod = top_prod.nlargest(10)

# Set the figure size for clarity
plt.figure(figsize=(9, 4))

# Plot a horizontal bar chart where x-axis is revenue in millions and  y-axis is
sns.barplot(
    x=top_prod.values,
    y=top_prod.index,
    palette='rocket'
)
# Add title and axis labels
plt.title('Top 10 Products by Revenue (in Millions)')
plt.xlabel('Total Revenue (in Millions)')
plt.ylabel('Products')

# Adjust layout to label overlap
plt.tight_layout()

# Display the plot
plt.show()
```

```
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ng:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(
```
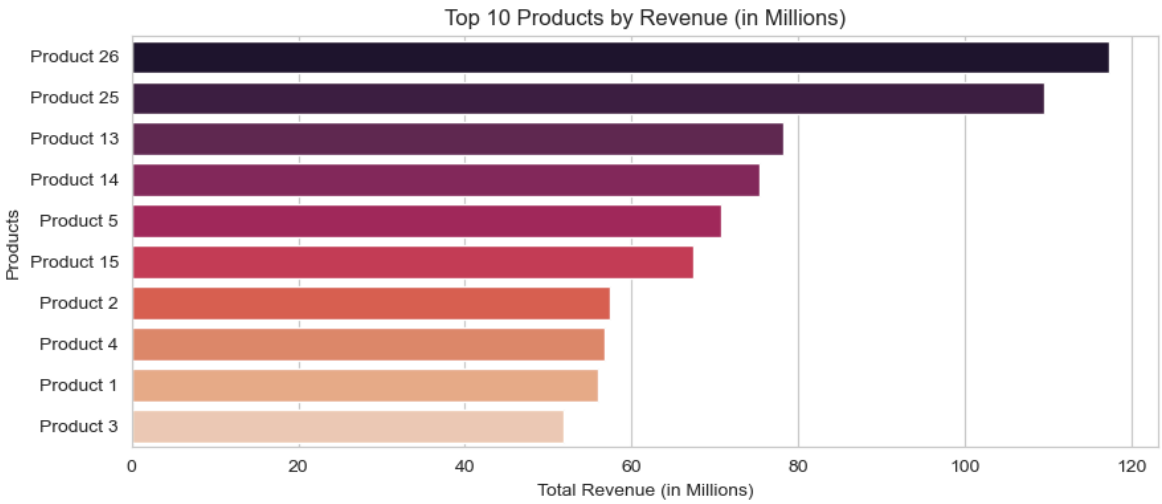



Insights---

Products 25 and 26 pull awau at $118Mand$110 M, with a sharp drop to $78Mforproduct13andatightmid-packat$68-$75M.

The bottom four cluster at $52-$57M, highlighting similar constraints.

Focus on growth pilots for the mid-tier and efficiency gains for the lower earners to drive significant lifts.

4.Top 10 products by avg profit margin

Goal: Compare average profitability across products to identify high-margin items

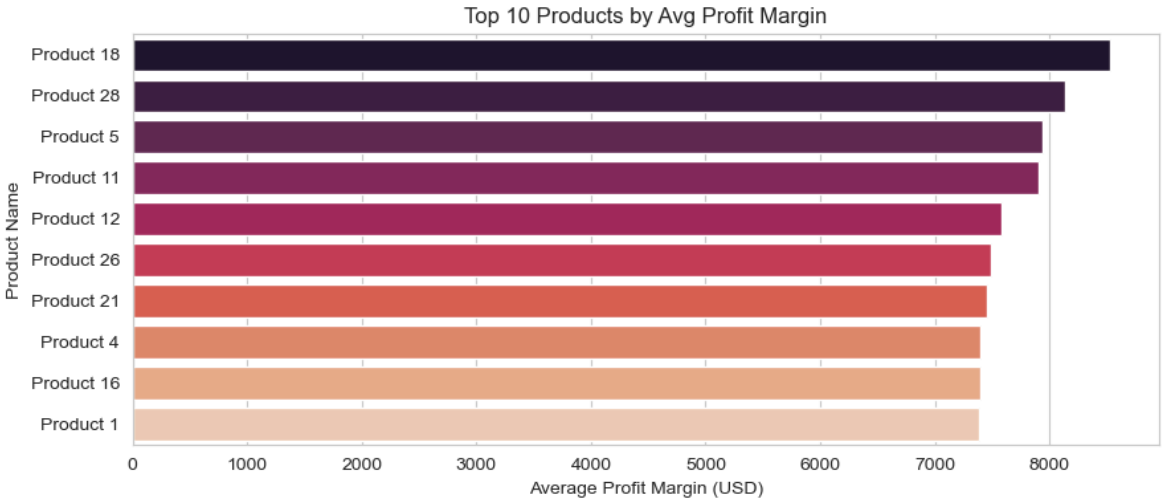
Chart: Horizontal bar chart

EDA Type: Univariate

Structure: Bars sorted descending to show top 10 products with average profit margin values

In [48]:
```
# 1.Compute average profit per product and take top 10
top_margin = (df.groupby('product_name')['profit']
                .mean()
                .sort_values(ascending=False)
                .head(10)
              )

# 2. Set figure size for clarity
plt.figure(figsize=(9, 4))

# 3 Plot a horizontal bar chart
sns.barplot(
    x=top_margin.values,
    y=top_margin.index,
    palette='rocket'
)

# 4. Add title and axis labels
plt.title('Top 10 Products by Avg Profit Margin')
plt.xlabel('Average Profit Margin (USD)')
plt.ylabel('Product Name')

# 5. Adjust layout to prevent label overlap
plt.tight_layout()

# 6. Display the plot
plt.show()
```

```
C:\Users\Mudit PC\AppData\Local\Temp\ipykernel_1468\332990340.py:12: FutureWarnin
g:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(
```



5. Sales by channel (Pie Chart)

Goal: Show distribution of total sales across channels to identify dominant sales routes

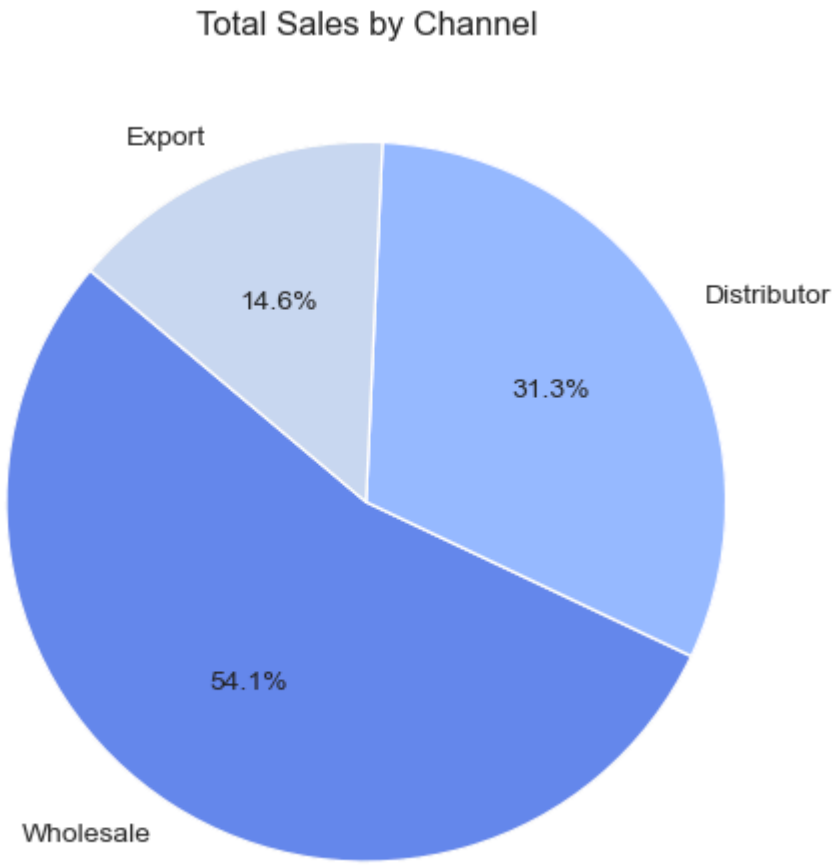
Chart: Pie Chart

EDA Type: Univariate

Structure: Pie segments with percentage labels, colors for clarity, start angle adjusted

In [49]:
```
# Group revenue by sales channel and sort descending
chan_sales = df.groupby('channel')[
    'revenue'].sum().sort_values(ascending=False)

# Set figure size for the pie chart
plt.figure(figsize=(5, 5))

# Plot pie chart with percentage labels and a defined start angle
plt.pie(
    chan_sales.values,
    labels=chan_sales.index,
    autopct='%1.1f%%',
    startangle=140,
    colors=sns.color_palette('coolwarm')
)

# Add title for context
plt.title('Total Sales by Channel')

# Adjust layout to ensure everything fits well
plt.tight_layout()
```

```
# Display the chart
plt.show()
```



Insights ---

Wholesale accounts for 54% of sales, with distributors at ~31% and exports at ~15%, underscoring reliance on domestic bulk channels.

To diversify revenue and mitigate concentration risk, prioritize expanding export initiatives - thorugh targeted overseas marketing and strategic partner relationships.

6. Average Order Value (AOV) Distribution

Goal: Understand distribution of order values to identify typical spending levels and outliers

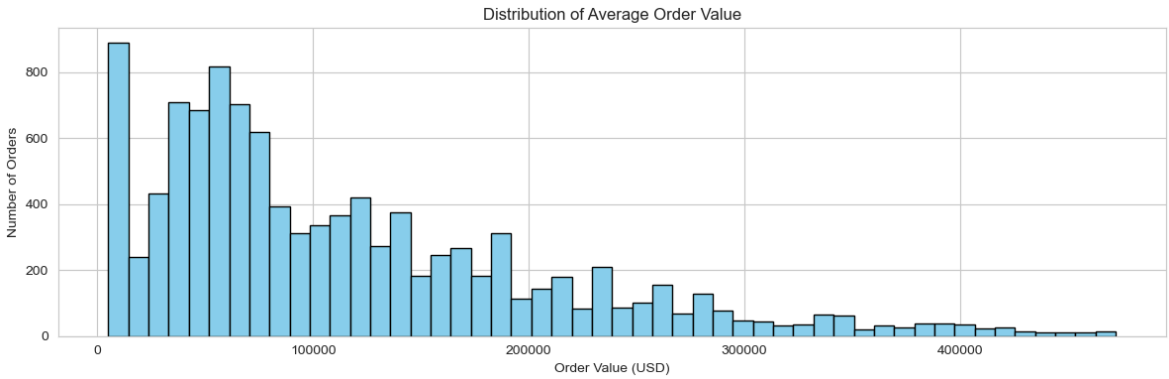
Chart: Histogram

EDA Type: Univariate

Structure: Histogram with 50 bins, colored bars with edge highlights to show frequency of order values.

In [50]:
```
# Calculate the total revenue for each order to get the order value
aov = df.groupby('order_number')['revenue'].sum()
```

```
# Set the figure size for better visibility
plt.figure(figsize=(12, 4))

# Plot a histogram of order values
plt.hist(
    aov,
    bins=50,
    color='skyblue',
    edgecolor='black'
)

# Add title and axis labels for content

plt.title('Distribution of Average Order Value')
plt.xlabel('Order Value (USD)')
plt.ylabel('Number of Orders')

# Adjust layout to prevent cliping
plt.tight_layout()

# Show the plot
plt.show()
```



Insights --

The order-value distribution is heavily right-skewed, with most orders clustering between $20K and$120 K and a pronounced mode around $50K-$60K. A Long tail of high-value transactions extends up toward $400K-$500K, but these large orders represent only a small share of total volume.

7. Profit Margin % vs. Unit Price

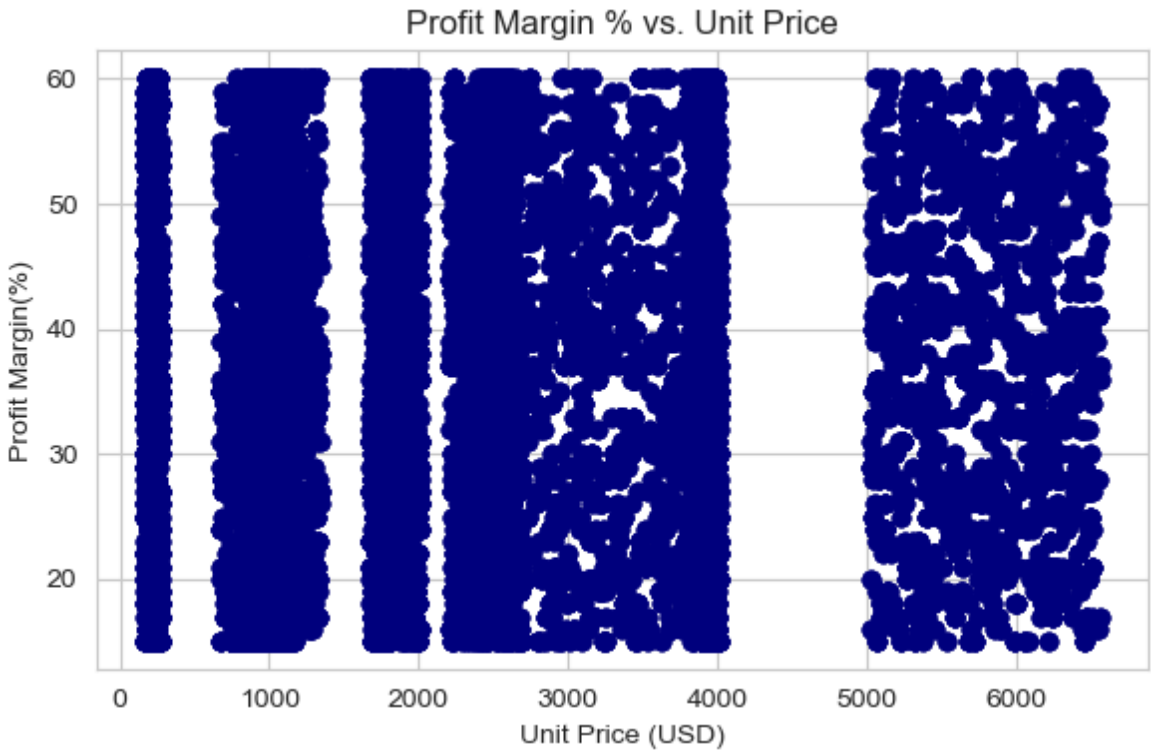
Goal: Examine relationship between unit price and profit margin percentage across orders

Chart: Scatter plot

EDA Type: Bivariate

Structure: Scatter points with transparency to show data density

In [51]:
```
# Calculate profit margin percentage for each record
df['profit_margin_pct'] = (df['profit']/df['revenue'])*100

# Set the figure size for clarity
plt.figure(figsize=(6, 4))

# Plot unit price vs. profit margin percentage
plt.scatter(
    df['unit_price'],
    df['profit_margin_pct'],
    alpha=0.6,
    color='navy'
)
# Adding titile and axis labels
plt.title('Profit Margin % vs. Unit Price')
plt.xlabel('Unit Price (USD)')
plt.ylabel('Profit Margin(%)')

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()
```



Insights ---

Profit margins are concentrated between ~18% and 60% with no clear correlation to unit price, which spans form near $0toover$6,500.

Dense horizontal bands indicate consitent margins tiers across a wide price spectrum, reflecting uniform pricing startegies.

Outliers below 18% at both low and high price points may signal cost inefficienceis or pricing issues worth deeper investigations.

8. Unit Price Distribution per Product

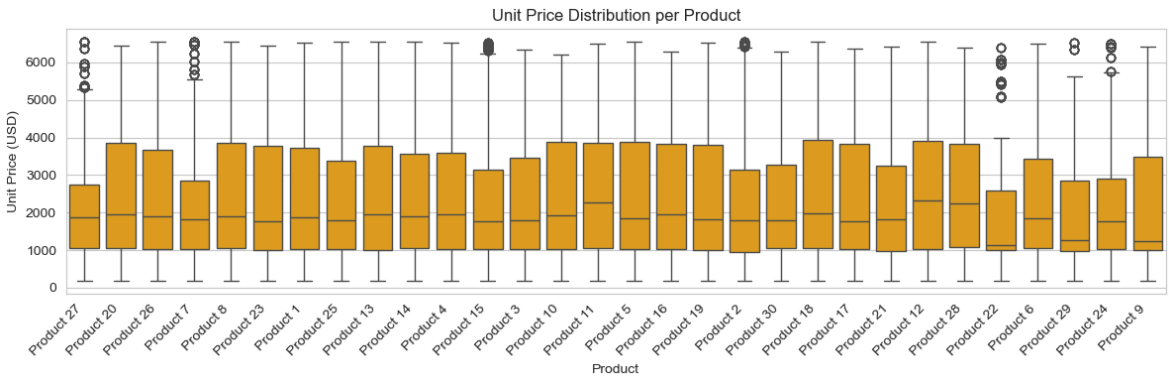
Goal: Compare pricing variability across different products to Identify price consistency and outliers

Chart:Boxplot

EDA Type:Bivariate

Structure:Boxplot with rotated labels to display unit price spread per product

In [52]:
```
# Set figure size for clarity
plt.figure(figsize=(12, 4))

# Create a boxplot of unit_price by product name

sns.boxplot(
    data=df,
    x='product_name',
    y='unit_price',
    color='orange'
)

# Adding title and lables
plt.title('Unit Price Distribution per Product')
plt.xlabel('Product')
plt.ylabel('Unit Price (USD)')

# Rotating x-axis labels for better readability
plt.xticks(rotation=45, ha='right')

# Adjust layout to prevent clipping of labels
plt.tight_layout()

# Display the plot
plt.show()
```



Insights ---

Products 8, 17, 27, 20 and 28 show high-end revenue spikes-well above their upper whiskers-likely due to bulk orders, special-edition releases, or premium bundles that temporarily inflate earnings.

In contrast, deep low-end outliers (near $0-100$) on Products 20 and 27 suggest promotional giveaways or test SKUs that pull down average prices.

To ensure accurate margin and pricing assessments, exclude these outlier transactions from average calculations.

9. Total Sales by US Region

Goal:Compare total sales across U.S. regions to identify top-performing markets and ares for target growth.

Chart:Horizontal bar chart

EDA Type: Univariate comparison

Structre:

1. Bars sorted ascending (Northeast -> West) for clear bottom-to-top ranking
2. X-axis in millions USD, Y-axis listing regions

In [53]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Aggregate total sales by region (in millions)
region_sales = (df.groupby('region')['revenue']
                .sum()
                .sort_values(ascending=False)
                / 1e6
                )

# 2. Plot
plt.figure(figsize=(10, 4))
sns.barplot(
    x=region_sales.values,
    y=region_sales.index,
    palette='rocket'
)

# 3. Formatting
plt.title('Total Sales by US Region', fontsize=16, pad=12)
plt.xlabel('Total Revenue (in Millions USD)', fontsize=12)
plt.ylabel('US Region', fontsize=12)
plt.xticks(rotation=0)
sns.despine(left=True, bottom=True)

plt.tight_layout()
plt.show()
```

```
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ng:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(
```

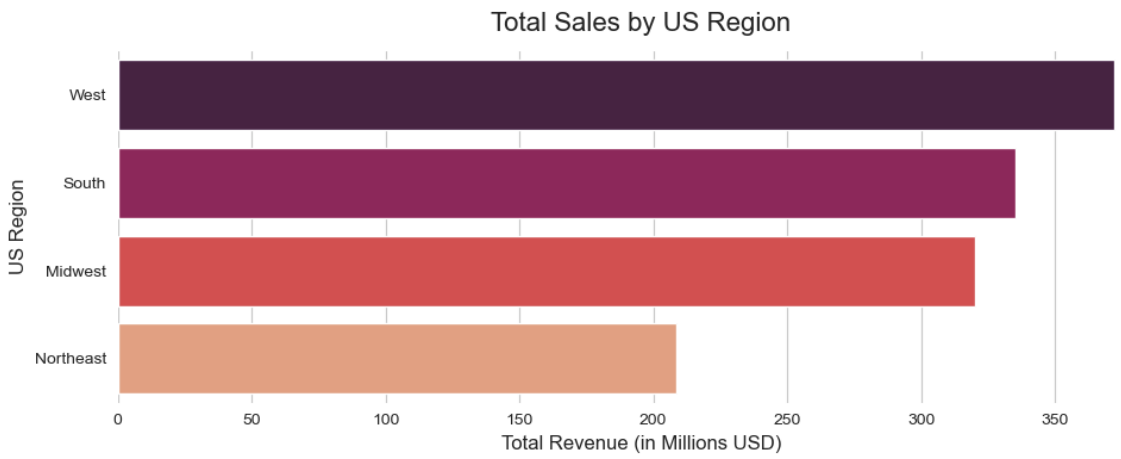

Regional Sales Insights

Regional Performance

- **West Region** dominates with approximately **$360M in sales (~35% of total revenue)**, highlighting strong market leadership.
- **South and Midwest** each contribute more than **$320M (~32%)**, indicating strong and consistent demand across central regions.
- **Northeast** generates roughly **$210M (~20%)**, suggesting potential room for growth and targeted market expansion.

Strategic Insight:
Focus on closing the **Northeast performance gap** through localized promotions, targeted marketing, and strategic partnerships while maintaining the successful national strategy used in higher-performing regions.

State-Level Performance

Top-performing state

- **California**
  - Revenue: ~**$230M**
  - Orders: **7,500+**

California significantly outperforms all other states, generating **more than twice the revenue of the next-highest market**.

Second-tier markets

- **Illinois**
- **Florida**
- **Texas**

These states each generate approximately $85M$–**110M in revenue** with **3,600 – 4,600 orders**, representing strong and stable regional demand.

Mid-tier markets

- **New York**
- **Indiana**

These markets produce around $54M$–**55M in revenue** with roughly **2,500 orders**.

Lower-performing markets

- **Connecticut**
- **Michigan**
- **Massachusetts**

Revenue ranges between $35M$–**44M** with approximately **1,600 – 2,100 orders**.

Order Value Insight

The close alignment between **revenue and order counts across states** suggests relatively consistent **average order values**.

Growth Opportunities

- Increase **order volume in mid-tier states**
- Improve **average order value in lower-performing regions**
- Expand marketing and distribution in underpenetrated markets

10. *Total Sales by State (Choropleth Map)*

Goal: Visualize geographic distribution of sales to identify high- and low-performing states and uncover regional gaps.

Chart: US choropleth map

EDA Type: Univariate geospatial

Structure:

- States shaded by total sales (in millions USD) using a blue gradient
- Legend on the right showing sales scale (M USD)
- Hover tooltips display exact sales for each state
- Map scoped to USA for clear regional context

In [54]:
```
import plotly.express as px

# 1. Aggregate total sales by region (in millions)
state_sales = (df.groupby('state')['revenue']
               .sum()
               .reset_index()
               )
state_sales['revenue_m'] = state_sales['revenue']/1e6
```

```
# 2. Plotly choropleth
fig = px.choropleth(
    state_sales,
    locations='state',
    locationmode='USA-states',
    scope='usa',
    color='revenue_m',
    color_continuous_scale='tealrose',
    labels={'revenue_m': 'Total Revenue (in Millions USD)'},
    hover_data={'revenue_m': ':,.2f'},
)

# 3. Layout tuning
fig.update_layout(
    title_text='Total Sales by State',
    margin=dict(l=0, r=0, t=40, b=0),
    coloraxis_colorbar=dict(
        title='Sales (M USD)',
        ticksuffix='M',
    )
)
fig.show()
```

INSIGHTS —

- **California leads** with $230\ M, followedbyIllinois$(112 M) and Florida ($90\ M), formingadominanttop\text{-}tier(>$90 M).
- **Mid-tier states** (e.g., Texas $85\ M, NewYork$55 M) hold steady performance but trail the top three by 40–145 M.
- **Lower-tier states** (e.g., New Jersey $47\ MdowntoMassachusetts$35 M) reveal a gradual drop, indicating uneven market penetration.
- **Action:** Double down on top states with tailored promotions, and launch targeted growth initiatives in under-penetrated regions to close the gap.

11. *Top 10 States by Revenue and Order Count*

Goal: Identify highest revenue-generating states and compare their order volumes

Chart: Two bar charts

EDA Type: Multivariate

Structure: First chart shows top 10 states by revenue (in millions), second shows top 10 states by number of orders

In [55]:
```
# 1 .Calculate total revenue per customer and select top 10
top_rev = (
    df.groupby('customer_name')['revenue']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
```

```python
# 2 .Caluculate total revenue per customer and select bottom 10
bottom_rev = (
    df.groupby('customer_name')['revenue']
    .sum()
    .sort_values(ascending=True)
    .head(10)
)
# 3 .Create a figure with two side-by-side subplots
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

# Plot 1: Top 10 customers by revenue (converted to millions)
sns.barplot(
    x=top_rev.values/1e6,
    y=top_rev.index,
    ax=axes[0],
    palette='magma'
)

# Plot 2: Bottom 10 customers by revenue (converted to millions)
sns.barplot(
    x=bottom_rev.values/1e6,
    y=bottom_rev.index,
    ax=axes[1],
    palette='magma'
)
axes[1].set_title('Bottom 10 Customers by Revenue', fontsize=14)
axes[1].set_xlabel('Revenue (Million USD)', fontsize=12)
axes[1].set_ylabel('Customer Name', fontsize=12)

# Adjust layout to prevent overlap and display both charts
plt.tight_layout()
plt.show()
```

```
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ng:



Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.


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ng:



Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.
```
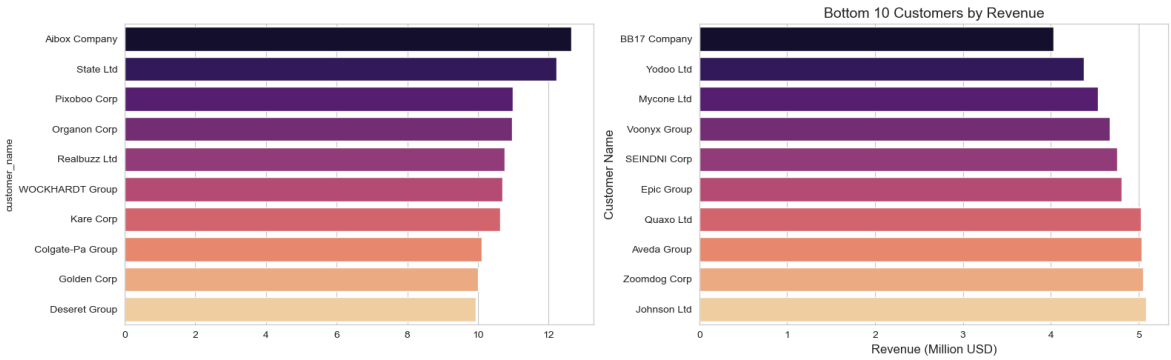



Insights ---

- Aibox Company tops the lists with $12.5M, followedcloselybystateltd(12.2$ M), while the 10th - ranked Desert Group still contributes $9.9M - demonstratingatight10$-12 M top tier.
- At the bottom, Johnson ltd leads its group with $5.1M, downtoBB17Companyat$ 4.1 M-roughly half the top customer's revenue.
- This steep drop from ~$10M + to4$-5 M highlights high revenue concentration among the top customers.
- Action: prioritize retention and upsell for your top ten, and launch targeted growth campaigns to elevate the lower-revenue cohort.

12. *Average Profit Margin by Channel*

Goal: Compare average profit margins across sales channels to identify the most and least profitable routes

Chart: Bar chart

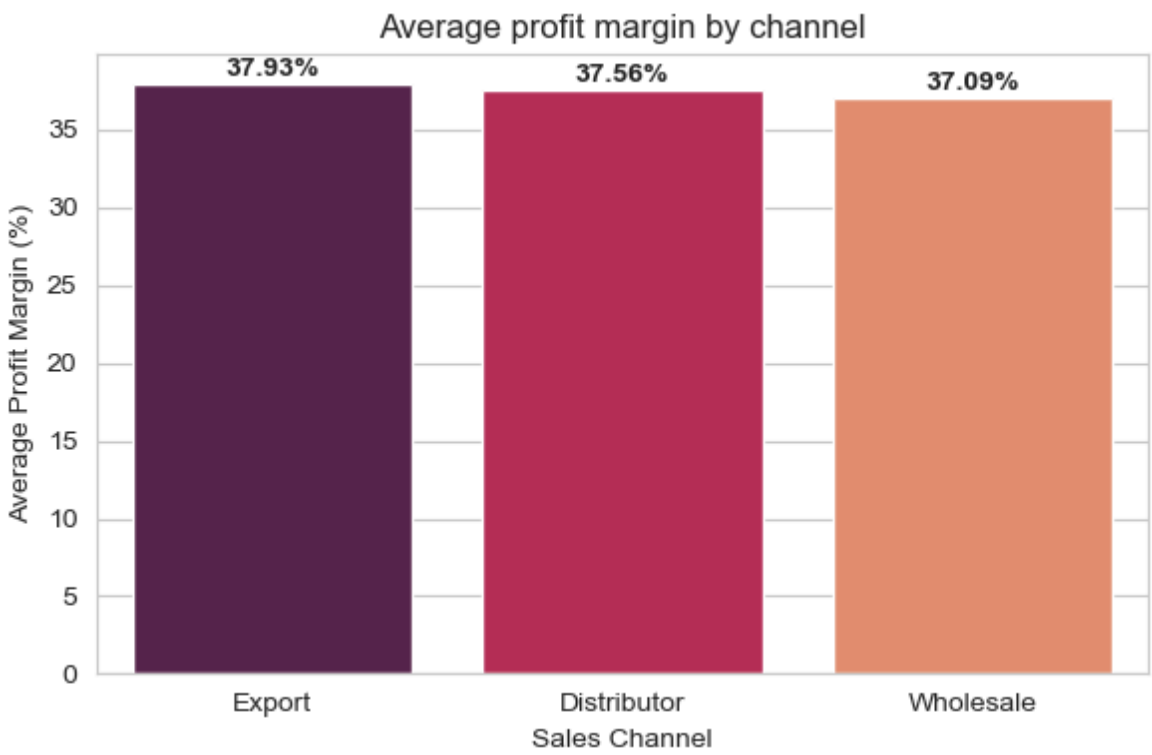
EDA Type: Bivariate

Structure: Vertical bars with data labels showing margin percentages, sorted descending by channel

In [56]:
```
# 1. Compute average profit margin percentage for each channel

channel_margin = (
    df.groupby('channel')['profit_margin_pct']
    .mean()
    .sort_values(ascending=False)
)
# 2. Set the figure size for clarity
plt.figure(figsize=(6, 4))

# 3. Plotting bar chart of average profit margin by channel
ax = sns.barplot(
    x=channel_margin.index,
    y=channel_margin.values,
```

```
    palette='rocket'
)

# 4. Add title and axis labels
plt.title('Average profit margin by channel')
plt.xlabel('Sales Channel')
plt.ylabel('Average Profit Margin (%)')

# 5. Annotate each bar with its exact margin percentage
for i, v in enumerate(channel_margin.values):
    ax.text(
        i,
        v+0.5,
        f"{v:.2f}%",
        ha='center',
        fontweight='bold'
    )
# 6 Adjust layout to prevent clipping and display the plot
plt.tight_layout()
plt.show()
```

```
C:\Users\Mudit PC\AppData\Local\Temp\ipykernel_1468\3611242263.py:12: FutureWarni
ng:



Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.
```



INSIGHTS ---

Export leads with a 37.93 % average margin, closely followed by Distributor (37.56 %) and Wholesale (37.09 %).

The tiny spread (<0.2 %) shows consistently strong profitability across all channels.

This uniformity implies well-controlled costs and pricing power everywhere.

To maximize returns, push volume growth in Export while maintaining efficiency in Distributor and Wholesale.

13. *Top and Bottom 10 Customers by Revenue*

Goal: Identify your highest- and lowest-revenue customers to tailor engagement strategies

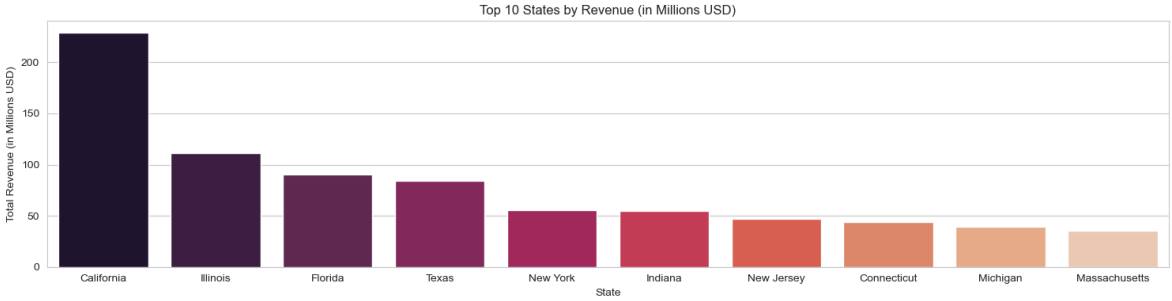
Chart: Side-by-side horizontal bar charts

EDA Type: Multivariate

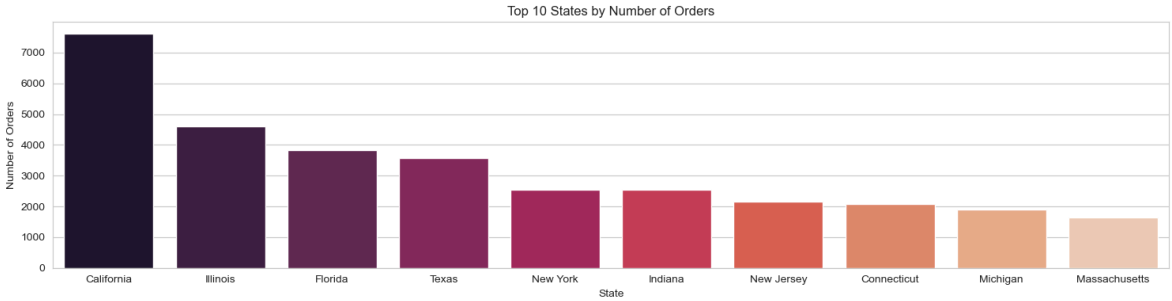
Structure: Left chart shows top 10 customers by revenue (in millions), right chart shows bottom 10 customers by revenue (in millions)

In [57]:
```
# 1. Aggregate total revenue ana unique order count per state
state_rev = df.groupby('state').agg(revenue=('revenue', 'sum'),
                                    orders=('order_number', 'nunique')
                                    ).sort_values('revenue', ascending=False).he

# Plot 1: Top 10 states by revenue (scaled to millions)
plt.figure(figsize=(15, 4))
sns.barplot(
    x=state_rev.index,
    y=state_rev['revenue']/1e6,
    palette='rocket'
)
plt.title('Top 10 States by Revenue (in Millions USD)')
plt.xlabel('State')
plt.ylabel('Total Revenue (in Millions USD)')
plt.tight_layout()
plt.show()

# Plot 2: Top 10 states by number of orders
plt.figure(figsize=(15, 4))
sns.barplot(
    x=state_rev.index,
    y=state_rev['orders'],
    palette='rocket'
)
plt.title('Top 10 States by Number of Orders')
plt.xlabel('State')
plt.ylabel('Number of Orders')
plt.tight_layout()
plt.show()
```

```
C:\Users\Mudit PC\AppData\Local\Temp\ipykernel_1468\575810476.py:8: FutureWarnin
g:


Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.
```




```
C:\Users\Mudit PC\AppData\Local\Temp\ipykernel_1468\575810476.py:21: FutureWarnin
g:


Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.
```



14. *Customer Segmentation: Revenue vs. Profit Margin*

Goal: Segment customers by total revenue and average profit margin, highlighting order volume

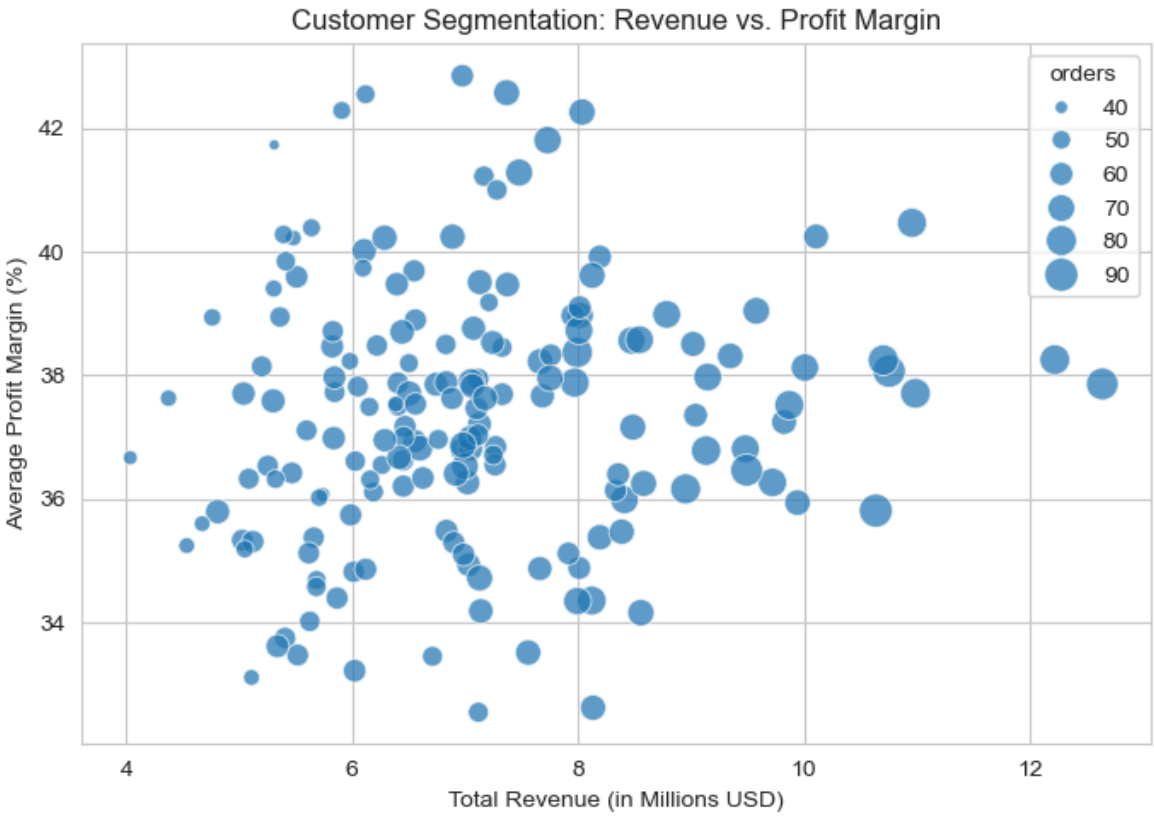
Chart: Bubble chart (scatter plot with variable point sizes)

EDA Type: Multivariate

Structure: Scatter points sized by number of orders, plotting revenue vs. margin

In [58]:
```
# 1 .Aggregate metrics per customer
cust_summary = df.groupby('customer_name').agg(
    total_revenue=('revenue', 'sum'),
    total_profit=('profit', 'sum'),
    avg_margin=('profit_margin_pct', 'mean'),
```

```
    orders=('order_number', 'nunique')
)

# Convert revenue to millions
cust_summary['total_revenue_m'] = cust_summary['total_revenue']/1e6
plt.figure(figsize=(7, 5))

# Convert revenue to million
sns.scatterplot(
    data=cust_summary,
    x='total_revenue_m',
    y='avg_margin',
    size='orders',
    sizes=(20, 200),
    alpha=0.7
)
plt.title('Customer Segmentation: Revenue vs. Profit Margin')
plt.xlabel('Total Revenue (in Millions USD)')
plt.ylabel('Average Profit Margin (%)')
plt.tight_layout()
plt.show()
```



INSIGHTS ---

- Customers with >$10 M in revenue tend to sustain margins between 36–40%, indicating that scale does not significantly erode profitability.
- Most customers cluster within the $6–10 M range and show stable margins (~34–40%), suggesting consistent pricing in this tier.
- Customers below $6 M display the widest margin variance (~33–43%), pointing to more volatile cost structures or discounts among smaller accounts.

- Bubble size (order count) increases with revenue, but margin levels appear unaffected—reinforcing revenue as the dominant performance driver over order volume.

15. *Correlation Heatmap of Numeric Features*

Goal: Identify relationships among key numeric variables to uncover potential multicollinearity

Chart: Correlation heatmap

EDA Type: Multivariate

Structure: Annotated heatmap with correlation coefficients for selected numeric columns

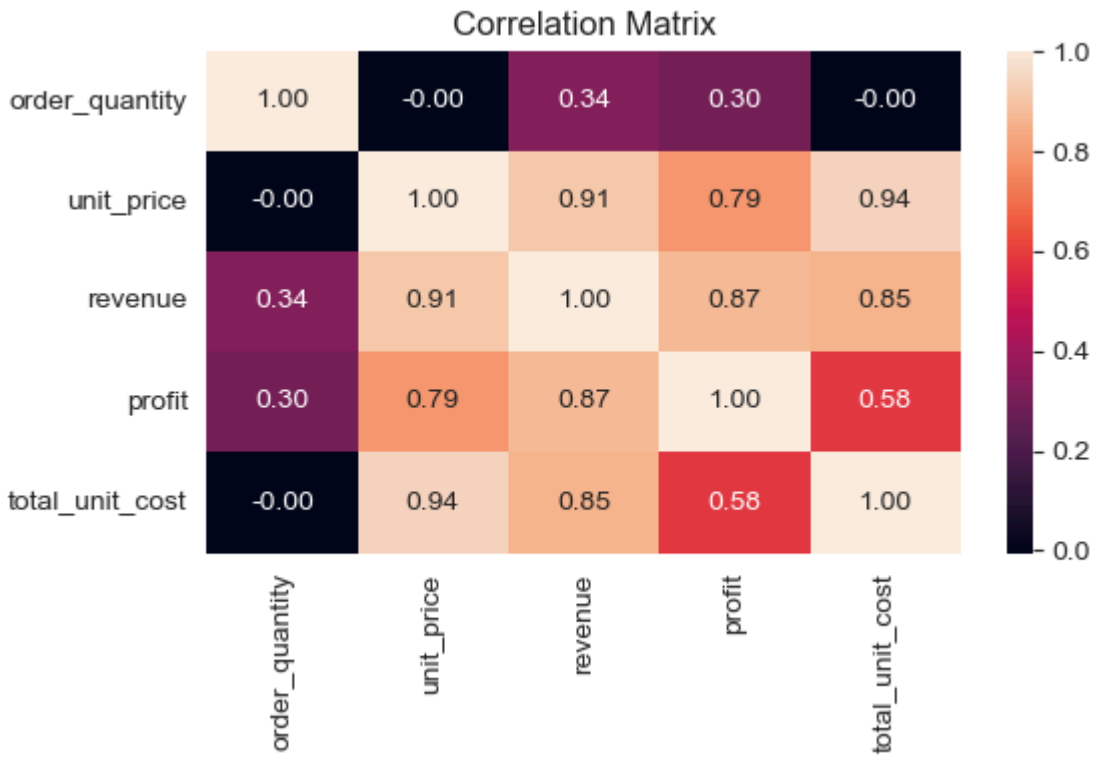
In [59]:
```
df.head(1)
```

Out[59]:

	order_number	order_date	customer_name	channel	product_name	order_quant
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	

1 rows × 23 columns

In [60]:
```
# List numeric columns to include in the correlation calculation
num_cols = ['order_quantity', 'unit_price',
            'revenue', 'profit', 'total_unit_cost']

# Calculate the correlation matrix for these numeric features
corr = df[num_cols].corr()

# Set the figure size for better visibility
plt.figure(figsize=(6, 4))

# Plot the heatmap with annotations and a custom color map
sns.heatmap(
    corr,
    annot=True,
    fmt='.2f',
    cmap='rocket'
)

# Add a title and show the plot
plt.title('Correlation Matrix')

# Adjust layout to prevent clipping
plt.tight_layout()

# Display the plot
plt.show()
```



INSIGHTS ---

- **Profit and revenue** are very strongly correlated (0.87), indicating that as sales value increases, profit tends to rise as well.
- **Unit price** is a key driver: it correlates 0.91 with revenue, 0.79 with profit, and 0.94 with cost—highlighting how pricing decisions ripple through both top-line and expense figures.
- **Cost** shows a strong link to revenue (0.85) but a more moderate tie to profit (0.58), underscoring that while higher sales often bring higher expenses, margins can still vary.
- **Quantity** has virtually no correlation with unit price or cost (≈0.00) and only modest associations with revenue (0.34) and profit (0.30), making volume a secondary factor compared to pricing.

Saving Pre-Processed Dataframe

In [61]:
```
df
```

Out[61]:

	order_number	order_date	customer_name	channel	product_name	order_
0	SO - 000225	2014-01-01	Rhynoodle Ltd	Wholesale	Product 27	
1	SO - 0003378	2014-01-01	Thoughtmix Ltd	Distributor	Product 20	
2	SO - 0005126	2014-01-01	Amerisourc Corp	Wholesale	Product 26	
3	SO - 0005614	2014-01-01	Colgate-Pa Group	Export	Product 7	
4	SO - 0005781	2014-01-01	Deseret Group	Wholesale	Product 8	
...	...	...	...	...	...	
64099	SO - 0007573	2018-02-28	Dazzlesphe Corp	Wholesale	Product 26	
64100	SO - 0007706	2018-02-28	Yombu Corp	Export	Product 21	
64101	SO - 0007718	2018-02-28	Bath Group	Distributor	Product 13	
64102	SO - 0008084	2018-02-28	Linklinks Ltd	Distributor	Product 20	
64103	SO - 0008654	2018-02-28	SAFEWAY Ltd	Distributor	Product 15	

64104 rows × 23 columns

In [62]:
```
# Export Final DataFrame to CSV
df.to_csv('mudit_sales_data_eda_completed_final.csv', index=False)
```

Key Insights

1. Revenue Stability and Monthly Trends

Revenue remains relatively stable between $23M and$**26.5M** during the 2014–2017 period, with no strong recurring seasonal spikes.

However, a noticeable decline occurs in **early 2017**, where revenue drops to approximately **$21.2M**, potentially indicating a **temporary market disruption or operational anomaly**.

---

2. Sales Channel Distribution

Channel	Share of Total Revenue
Wholesale	54%
Distributors	31%
Export	15%

Sales are heavily concentrated in the **Wholesale channel**, suggesting a strong reliance on bulk distribution. The relatively smaller export share indicates a **potential opportunity to expand international markets**.

---

3. Top Revenue-Generating Products

Top performers

- **Product 26:** $118M
- **Product 25:** $110M
- **Product 13:** $78M

Mid-tier performers

- Products generating approximately $68M–$**75M**

Lower-performing products

- Products generating approximately $52M–$**57M**

This distribution indicates that **a small number of products contribute a significant share of total revenue**.

---

4. Profit Margin Distribution

Profit margins range broadly from approximately **18% to 60%**.

Key observations:

- No strong correlation between **unit price and profit margin**
- Horizontal clustering suggests **standardized pricing strategies across product tiers**

---

5. Seasonal Sales Volume

Monthly sales volumes show **limited seasonal variation**, although a slight increase in demand appears during **May–June**.

The **early 2017 revenue dip (~$21.2M)** stands out as an anomaly and may require further investigation.

---

6. Regional Sales Performance

Top-performing market

- **California**
  - Revenue: ~**$230M**
  - Orders: **7500+**

Mid-tier markets

- **Illinois, Florida, Texas**

  - Revenue: $85M-$**110M**
  - Orders: **3500 – 4500**

Lower-volume markets

- **New York, Indiana**
  - Revenue: ~**$54M**
  - Orders: **2000+**

These patterns highlight **regional demand concentration**, with California acting as the primary revenue driver.

Recommendations

1. **Outlier Strategy:** Exclude or formalize bulk-order and promotional SKUs when calculating averages.
2. **Margin Uplift:** Apply top-product pricing levers to mid/low tiers; cut costs on underperformers.
3. **Export Growth:** Invest in targeted overseas marketing and distributor partnerships.
4. **Seasonal Planning:** Shift spend toward January trough and May–June peak; investigate the 2017 anomaly.
5. **Dashboard Prep:** Build aggregated tables for time series, channel mix, and product performance for Power BI.